

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Estimating Gradients

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ESTIMATING GRADIENTS · CH₃ · L6



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The Map

This lesson collapses Estimating Gradients into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw a tangent

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Use a large gradient triangle

TRIGGER *"gradient", "tangent", "rate of change"*

03 Estimate average gradient

TRIGGER *"gradient", "tangent", "rate of change"*

04 Interpret the rate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

06 Draw a tangent to the

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BANK EVIDENCE 3 website questions, 3 checked solutions, 3 local PDFs read.

01 Draw a tangent

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6 , tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. Use the graph to find the two possible values of k .

YOUR SOLUTION

02 Use a large gradient triangle

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

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Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. (2) Use the graph to find the two possible values of k.

YOUR SOLUTION

03 Estimate average gradient

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

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WORKED MODEL

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Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. Use the graph to find the two possible values of k .

YOUR SOLUTION

04 Interpret the rate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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FIRST LINE TO WRITE

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WORKED MODEL

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Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. (2) Use the graph to find the two possible values of k.

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

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Practice 05

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QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. Use the graph to find the two possible values of k .

YOUR SOLUTION

06 Draw a tangent to the

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

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Finish: Gradient ≈ -0.6 , tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

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Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. Use the graph to find the two possible values of k .

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw a tangent
"gradient", "tangent", "rate of change"	Use a large gradient triangle
"gradient", "tangent", "rate of change"	Estimate average gradient
"work out", "show", "hence", a familiar exam phrase	Interpret the rate
"work out", "find", a missing value	Find the gradient
"draw", "plot", "construct", a dia- gram/grid	Draw a tangent to the

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.